JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SREE VARI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

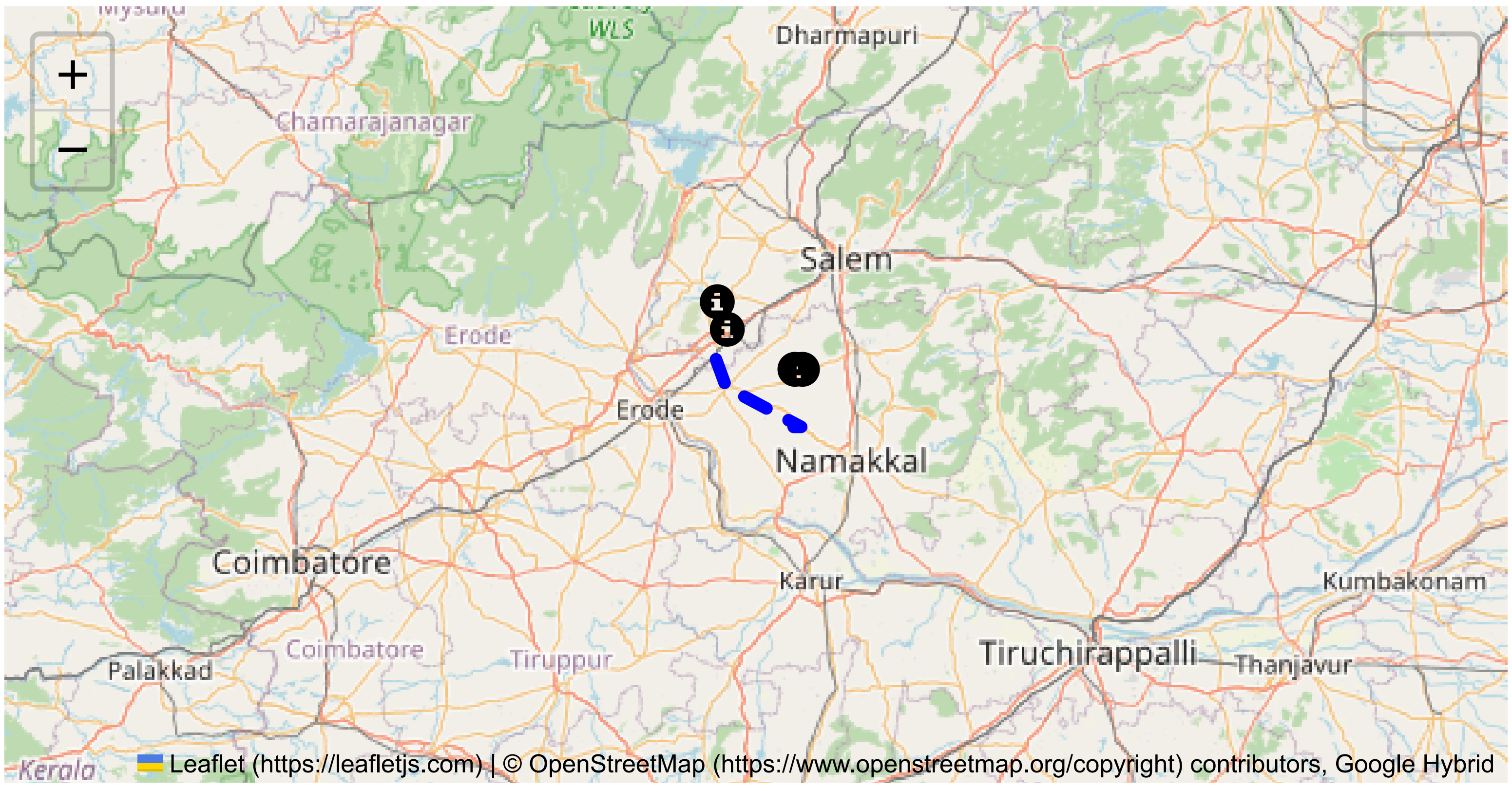
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

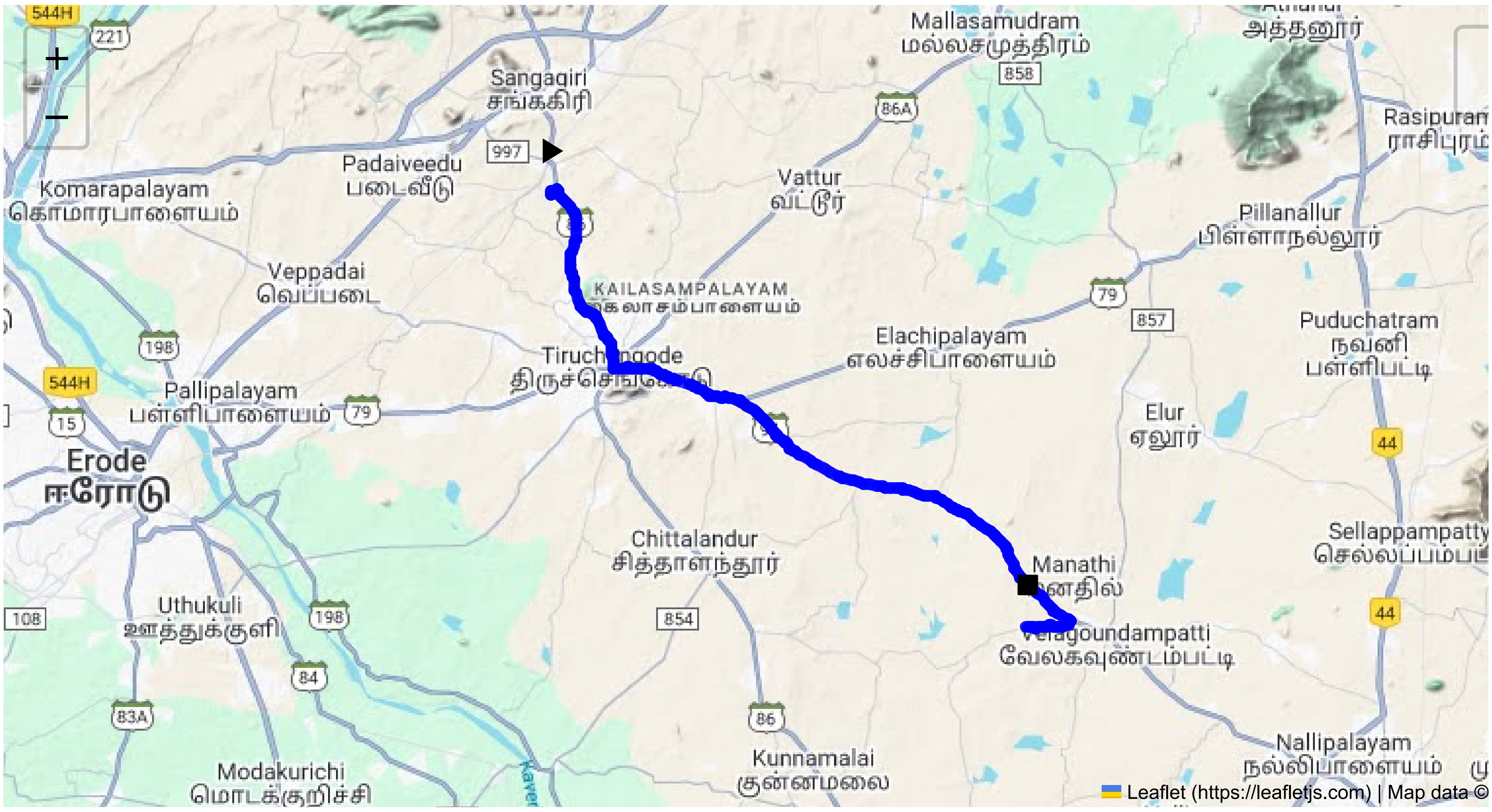
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 31.07 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours
Start: (11.4381, 77.8734)
End: (11.28891, 78.03998)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 72QR+PCJ, Thaligai spans approximately 31.07 kilometers. It passes through several key areas, including S Car St in Kamalar, Tiruchengode, and Marukalampatti. The road typically includes a mix of state highways and local roads, featuring rural and semi-urban settings.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. The potential weather-related hazards include:

- **Monsoon season**: Heavy rains from October to December can lead to road flooding and reduced visibility.
- **Summer heat**: From March to June, extreme heat can affect vehicle performance. Overall, the driver should be prepared for sudden weather changes during monsoon and follow weather updates closely.

3. Analysis of Traffic Patterns

- **Peak Hours**: Typically, congestion is higher during the morning (8-10 AM) and evening (5-7 PM) times due to local commute and school runs.
- **Congestion-Prone Areas**: Tiruchengode, being a more urbanized center, might experience mild congestion around market areas. Major junctions and intersections near towns may also hold up traffic, especially during peak hours.

4. Assessment of Road Quality and Infrastructure

- **General Road Condition**: The condition varies with main highways well-maintained, while rural sections may have uneven surfaces and potholes.
- **Infrastructure Concerns**: Watch for narrow bridges and roads, especially on regional roads. Road signage might be inconsistent in less urbanized areas.

5. Suggestions for Alternative Routes for Emergencies

In the event of road closures or significant delays:

- **Route 1**: Consider taking State Highway 95, bypassing Tiruchengode, which may offer a less congested albeit slightly longer road surface.
- **Route 2**: NH44 (National Highway) offers an emergency thoroughfare if primary roads are compromised.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Truck drivers are required to adhere to Tamil Nadu State Transport Department regulations for hazardous materials. This includes specific signage, permits, and operating hours for large vehicles. Ensure compliance with no-entry zones during peak hours in urban centers.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While this specific route hasn't recorded major incidents, areas with increased traffic, such as Tiruchengode, have seen vehicular mishaps due to congestion and narrow roads. Drivers should exercise extra caution.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas**: Avoid disposal or leakage of hazardous materials near agricultural lands and water bodies along the route.
- **Wildlife Movement**: Be aware of potential wildlife crossings, particularly in rural areas.

9. Analysis of Communication Coverage

Mobile network coverage is generally good along the route but may experience dead zones in rural or dense forested areas. Providers such as Airtel and Jio offer extensive coverage but verify carrier strength before departure.

10. Estimated Emergency Response Times for Different Route Segments

- **Urban and Semi-urban Areas (Tiruchengode)**: Rapid response within 20-30 minutes due to proximity to emergency services.
- **Rural Sections (Marukalampatti)**: Expect slower response times, potentially 30-60 minutes, due to lower service density.

11. Overall Summary of Risk Assessment

- **High Risks**: Weather-related disruptions in monsoon, road quality issues.
- **Moderate Risks**: Traffic congestion, especially in urban areas, accident-prone zones due to narrow roads.
- **Low Risks**: Communication issues are minimal, though some areas may have weak signals.

Conclusion: The route involves moderate risk factors primarily related to weather, road quality, and traffic patterns. With adherence to local regulations and proactive monitoring of weather and traffic conditions, the journey can be managed effectively. Drivers should plan for potential emergencies, alternative routes, and ensure that all regulatory requirements are strictly followed for safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Blind Spot	Blind Spot	11.29183, 78.05599	10 KM/Hr
7	Turn	Medium	11.28990, 78.05311	30 KM/Hr
8	Turn	Medium	11.29044, 78.04863	30 KM/Hr
9	Turn	Medium	11.29043, 78.04848	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
11	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
13	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
9	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
10	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
12	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.29183, 78.05599

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.28990, 78.05311

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.29044, 78.04863

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.29043, 78.04848